ZEERBOX · USER REQUIREMENTS

ZeerBox user requirements

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First issue (kit 1.9.1 upgrade), drawn from the users, operating environment and constraints in ZBX-PRB-001 v0.8

ZeerBox user requirements

What smallholder growers, farmer groups, traders and the local builders and technicians who would build and mend ZeerBox need, written from their side. No user has been consulted yet: every row comes from ZBX-PRB-001 (users table, operating environment and constraints) and from the published field work it cites, and each is to be checked in co-design sessions with the first candidate partner (MIT D-Lab's evaporative cooling team, a dry-season vegetable area of Mali; ZBX-DEC-001). The product requirements that meet these needs are in ZBX-PRS-001 (`03-requirements.md`), and the links are in the generated traceability matrix ZBX-RTM-001 (`03b-traceability.md`). Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Hold several days of harvest	I can keep one to five days of picking, about 20 crates, so I sell when the price is right and not on the day I pick	High	Smallholder vegetable grower; farmer group (PRB Table 1)
UR-002	Usability	Load without heavy lifting	I can walk in and reach every crate without lifting a full crate above my shoulders or bending into a pit	Medium	Smallholder grower; market trader (PRB, gap 2: ground-level passive coolers)
UR-003	Performance	Noticeably cooler on hot dry days	On a hot, dry afternoon the produce sits clearly cooler than the air outside	High	Smallholder grower (PRB, design climate and gap 1)
UR-004	Performance	Produce does not wilt	Leafy greens and tomatoes do not dry out and shrivel while stored	High	Smallholder grower; market trader (PRB, produce conditions)
UR-005	Usability	Know when it is not cooling	I can see at a glance whether the store is cooling, airing or idle, so I know when to keep the door shut and when the store is not helping	High	Smallholder grower; farmer group (PRB, gap 3)
UR-006	Safety	No mold and safe air	The store does not grow mold or smell, and the air inside is safe to walk into	High	Farmer group; market trader (PRB, Safety)
UR-007	Performance	Produce lasts longer	Tomatoes and peppers keep noticeably longer than in the shade outside	High	Smallholder grower (PRB, the problem)
UR-008	Logistics	Works without the grid	It runs where there is no grid or an unreliable one	High	All users (PRB, operating environment: power)
UR-009	Logistics	Water I can carry	The water it needs on the hottest day is an amount I can carry or pump by hand, and it holds a day's supply	Medium	Smallholder grower; farmer group (PRB, operating environment: water)
UR-010	Safety	Safe to touch and work on	Nothing electrical can give a shock, the fans cannot catch a hand and nobody can be shut in	High	All users; local electrician (PRB, constraints)
UR-011	Deployment and assembly	Built by local trades	A local mason and carpenter can build it from bricks, timber and sheet they already buy, with no refrigeration trade	High	Local builder (PRB Table 1 and constraints)
UR-012	Maintenance	Mended from the regional town	A local electrician can swap a fan, pump, sensor or pad quickly with a generic part from the regional town	High	Local electrician or phone-repair technician (PRB Table 1)
UR-013	Environment and durability	Lasts in heat, dust, termites and wind	The structure lasts many years, and the moving parts several seasons, in heat, dust, termites and strong wind	Medium	Farmer group; local builder (PRB, site)

ID	Category	Title	Need	Priority	Source
UR-014	Cost	Affordable to a group	A farmer group can afford the cooling equipment, and the building is priced in local materials	High	Farmer group or cooperative (PRB, constraints)

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.

Regulatory needs (food safety practice, battery and solar parts) are in the regulatory register ZBX-REG-001; they are not written as user requirements because no user stated them.